adk-based_Agent — Google ADK Agent Project Collection

This project directory contains standard agent packages built with the **Google Agent Development Kit (ADK)**. Each agent is structured as an independent Python package following ADK specification, ready to run and test via terminal CLI, web GUI, or programmatic Runners.

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1. Included Agent Folders

Folder	Agent Role	Core Architecture & Tools	Key Learning Points
`chat_time_weather_agent`	World city time & weather assistant and general chatter	Open-Meteo API integration, `geocode`, `get_current_time`, `get_weather`, offline fallback DB	• Tool auto-routing vs direct LLM response branch • IANA Timezone and WMO weather code resolution • PEP 8 & `AgentLoader` compliant folder design
`memory_layers_agent`	Dual-layer memory (Short-term Session vs Long-term Memory) assistant	Session State (`user:name`, `trip`), Memory Service (`preload_memory`, `load_memory`)	• Session State prefix (`user:`, `temp:`, unqualified) scopes • Prompt injection via `{key?}` template substitution • Cross-session persistent knowledge accumulation & retrieval

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2. Directory Structure

```
adk-based_Agent/
├── chat_time_weather_agent/          # Time, weather, and general conversation agent
│   ├── __init__.py                  # Exposes agent module and root_agent
│   ├── agent.py                     # Tool implementations and agent definition
│   ├── README-KR.md                 # Korean detailed documentation
│   └── README-EN.md                 # English detailed documentation
├── memory_layers_agent/             # Dual-layer memory agent
│   ├── __init__.py                  # Exposes agent module and root_agent
│   ├── agent.py                     # State tools and root_agent definition
│   ├── README-KR.md                 # Korean detailed documentation
│   └── README-EN.md                 # English detailed documentation
├── README-KR.md                     # Comprehensive Korean guide for the project
└── README-EN.md                     # Comprehensive English guide (this document)
```

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3. Prerequisites and Configuration

1) Package Installation

Install Google ADK and required libraries in a Python 3.10+ environment:

```
pip install google-adk python-dotenv
# or using uv
uv pip install google-adk python-dotenv
```

2) Environment Variables (`.env`) Configuration

Create a `.env` file in the parent folder (`part02_architecture` or workspace root):

```
# Gemini Developer API Key (Default)
GEMINI_API_KEY=your_gemini_api_key_here

# Or when using Google Cloud Vertex AI
# GOOGLE_GENAI_USE_VERTEXAI=TRUE
# GOOGLE_CLOUD_PROJECT=your-gcp-project-id
# GOOGLE_CLOUD_LOCATION=us-central1

# Default Agent model (Optional)
ADK_MODEL=gemini-flash-latest

# Offline testing mode (0: online API, 1: built-in offline data)
AGENT_OFFLINE=0
```

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4. Execution and Testing Methods

The agents in this project can be executed at three different levels:

Method 1: ADK CLI Interactive Mode (`adk run`)

Chat directly with the agents in your terminal:

```
# Run from within the adk-based_Agent directory

# 1) Run Time & Weather Agent
adk run chat_time_weather_agent

# 2) Run Dual-Layer Memory Agent
adk run memory_layers_agent
```

Method 2: ADK Web Development UI (`adk web`)

Test visually in your browser using Google ADK's official web UI:

```
# Launch web server from within the adk-based_Agent directory
adk web --port 8008
```

Open `http://localhost:8008` in your browser, select your target agent (`chat_time_weather_agent` or `memory_layers_agent`) from the top-left dropdown, and start chatting.

Method 3: Direct_ADK Integration Scripts

Observe internal event lifecycles and memory transitions in code using the test scripts located in the parent directory:

```
# Navigate to part02_architecture directory
cd ..

# 1) Observe request event lifecycle for chat_time_weather_agent
uv run Direct_ADK/run_events.py

# 2) Run dual-layer memory tests for memory_layers_agent (offline structure / Gemini integration)
uv run Direct_ADK/run_memory_layers.py offline
uv run Direct_ADK/run_memory_layers.py llm
uv run Direct_ADK/run_memory_layers.py all
```

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5. Detailed Documentation Links

For detailed tool signatures, instruction prompt engineering, dialogue scenarios, and troubleshooting, please refer to the documentation in each folder:

- chat_time_weather_agent Guide (Korean: README-KR.md) / English (README-EN.md)
- memory_layers_agent Guide (Korean: README-KR.md) / English (README-EN.md)